TrES-1b

R-1796

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-1_140901` · created 2026-08-03T01:27:20+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456901.6713 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1419 +/- 0.007
Transit depth (Rp/Rs)^2	2.01 +/- 0.2 [%]
Orbital Inclination (inc)	89.78 +/- 0.55
Airmass coefficient 1 (a1)	1.1617 +/- 0.0076
Airmass coefficient 2 (a2)	-0.1214 +/- 0.0071
Scatter in the residuals of the lightcurve fit is	0.65 %
Best Comparison Star	None
Optimal Aperture	3.81
Optimal Annulus	7.07
Transit Duration (day)	0.0951 +/- 0.0012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1419 ± 0.007 vs 0.1358 (deviation 0.87σ)
Duration fit vs published	0.0951 d vs 0.1042 d (deviation 7.56σ)
Mid-time O–C	-2.5 min
Depth SNR	20.3
Residual scatter	0.65 %

Light curve

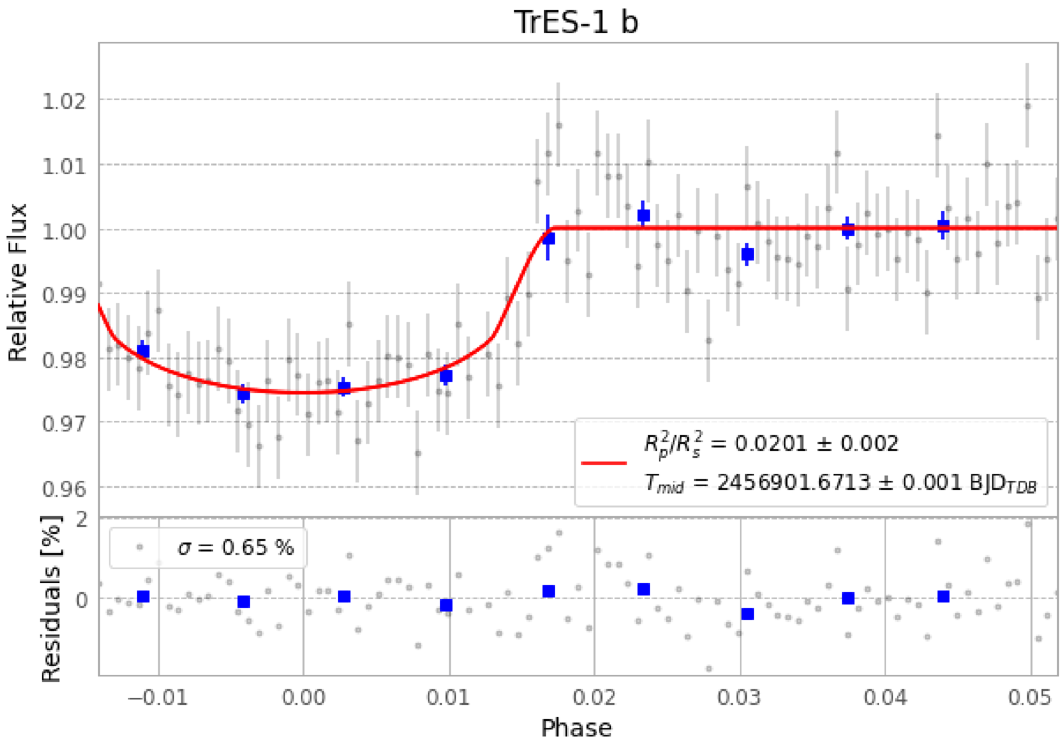

Figure 1 — FinalLightCurve_TrES-1 b_2014-08-31.png (SHA-256 9810b77fa5ef9c5b…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [208, 256], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 049052babab1a40e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

97 FITS frames (64 MB), `TRES-1140901030012.FITS` → `TRES-1140901074813.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-140901.log (966712c1e4d1…), FinalLightCurve_TrES-1 b_2014-08-31.png (9810b77fa5ef…), FinalLightCurve_TrES-1 b_2014-08-31.csv (9f7e5a99a386…)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)